ENGINEERING REFERENCE

Full-Stack & AI Infrastructure Specification

Complete inventory and architectural classification of production technologies, comprising frontend applications, inference engines, vector retrieval pipelines, and on-premise compute infrastructure.

APPLICATION LAYER & FRONTEND

LOGO	TECHNOLOGY	FUNCTION	ROLE & ARCHITECTURAL CONTEXT
	Next.js	Framework	Full-stack React platform for SSR, API routes, and static site generation.
	React	UI	Component library driving interactive client interfaces and state hierarchy.
	TypeScript	Language	Static typing across frontend components, state schemas, and API contracts.
	Tailwind	Styling	Utility-first styling system governing responsive, tokenized design elements.

BACKEND SERVICES & PERSISTENCE

LOGO	TECHNOLOGY	FUNCTION	ROLE & ARCHITECTURAL CONTEXT
	Python	Language	Primary language powering inference microservices, model pipelines, and scripts.
	FastAPI	API	Asynchronous REST and streaming gateway interface for inference requests.
	Node.js	Runtime	Server runtime handling edge SSR functions and auxiliary backend middleware.
	PostgreSQL	Database	ACID-compliant relational store for tenant entities, accounts, and session data.
	Redis	Cache	In-memory caching tier, rate limiting, and ephemeral session storage.

MACHINE LEARNING & RETRIEVAL (AI/LLM)

LOGO	TECHNOLOGY	FUNCTION	ROLE & ARCHITECTURAL CONTEXT
	PyTorch	Training	Deep learning library for fine-tuning, embedding generation, and experimentation.
	vLLM	Serving	High-throughput LLM serving engine with PagedAttention and continuous batching.
	pgvector	Search	Vector similarity indexing (HNSW / IVFFlat) embedded directly in PostgreSQL.
	RAG pipelines	Retrieval	Document ingestion, semantic chunking, re-ranking, and context synthesis.

INFRASTRUCTURE, NETWORKING & PHYSICAL COMPUTE

LOGO	TECHNOLOGY	FUNCTION	ROLE & ARCHITECTURAL CONTEXT
	Cisco	Networking	Core enterprise switching, inter-VLAN routing, and hardware security gateways.
	Ubiquiti	Wi-Fi	Enterprise wireless access points, management controllers, and branch links.
	Structured cabling	Cabling	Cat6A/fiber patch topology providing low-latency backbone interconnects.
	GPU racks	Compute	High-density dedicated GPU enclosures hosting accelerated inference workloads.
	AWS	Cloud	Public cloud bursting, object storage (S3), and managed failover services.
	Cloudflare	Edge	Global CDN, reverse proxy, DDoS mitigation, and edge DNS management.
	Docker	Containers	Standard containerization images for reproducible build and run environments.
	Linux	OS	Host operating system kernel orchestrating bare-metal and container processes.

> **Architecture Summary:** This stack pairs a modern web and API delivery tier with an on-premise high-density AI cluster (vLLM on GPU racks over low-latency Cisco fabric), complemented by Cloudflare edge caching and AWS cloud storage.